Credit Card Approval Prediction Using Machine Learning

ML-Powered Credit Card Approval System

Repository: github.com/ksangeeth9391/Credit-Card-Approval-Prediction

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Abstract

The Credit Card Approval Prediction System is a machine learning-based web application that predicts whether a customer's credit card application will be approved or rejected. The system analyzes applicant details such as income, employment status, education level, family status, housing type, and other financial information. Multiple classification algorithms are trained and evaluated to identify the best-performing model. The selected model is integrated into a Flask web application, allowing users to enter applicant details and receive instant prediction results. This project reduces manual effort, improves consistency in decision-making, and demonstrates the practical application of machine learning in the banking sector.

Problem Statement

Financial institutions receive thousands of credit card applications every day. Manual verification of each application is time-consuming, error-prone, and inconsistent. Banks require an intelligent system that can automatically analyze applicant information and accurately predict whether a credit card application should be approved or rejected. This project addresses that problem using machine learning techniques to automate the approval process and support faster, data-driven decision-making.

Features

- User-friendly Flask web application
- Applicant data input form
- Automatic data preprocessing (cleaning, encoding, scaling)
- Machine Learning-based prediction
- Multiple classification algorithms compared
- Best model selection based on accuracy
- Instant approval / rejection result
- Responsive HTML/CSS interface

Tech Stack

Category	Technology
Programming Language	Python
IDE for ML	Jupyter Notebook
IDE for Web	Visual Studio Code
Backend	Flask
Frontend	HTML, CSS
Machine Learning	Scikit-learn, XGBoost
Data Analysis	Pandas, NumPy

Visualization	Matplotlib, Seaborn
Model Storage	Joblib (Pickle-based)
Version Control	Git & GitHub

Algorithms Used

- Logistic Regression
- Decision Tree ✅ (best-performing model, selected for deployment)
- Random Forest
- XGBoost Classifier

Pre-requisites

- Python programming proficiency (Python 3.x fundamentals)
- Familiarity with classification algorithms (Logistic Regression, Decision Tree, Random Forest, XGBoost)
- Basic Flask knowledge (routing, templates, form handling)
- HTML/CSS skills for building the interface
- Working knowledge of Pandas, NumPy, Matplotlib, and Scikit-learn
- Git/GitHub for version control

Installation Steps

Step 1 — Clone the repository

```
git clone https://github.com/ksangeeth9391/Credit-Card-Approval-Prediction.git
```

Step 2 — Move into the project folder

```
cd Credit-Card-Approval-Prediction
```

Step 3 — Install required libraries

```
pip install -r requirements.txt
```

Step 4 — Run the Flask application

```
python app.py
```

Step 5 — Open the browser

```
http://127.0.0.1:5000
```

Usage Instructions

1. Launch the Flask application.
2. Open the web application in a browser.

3. Enter applicant details.
4. Click the Predict button.
5. The system processes the input using the trained machine learning model.
6. The prediction result (Approved or Rejected) is displayed instantly.

Deployment

The application is deployed and publicly accessible on Render.

Live URL: https://credit-card-approval-prediction-4-hk05.onrender.com

Screenshots

Applicant Information Form

Two-panel glassmorphism interface — an introduction panel on the left and the applicant details form (Gender, Own Car, Own House, Income, Education, Family Status, Housing Type, Age, Employment Years, Occupation, contact flags) on the right.

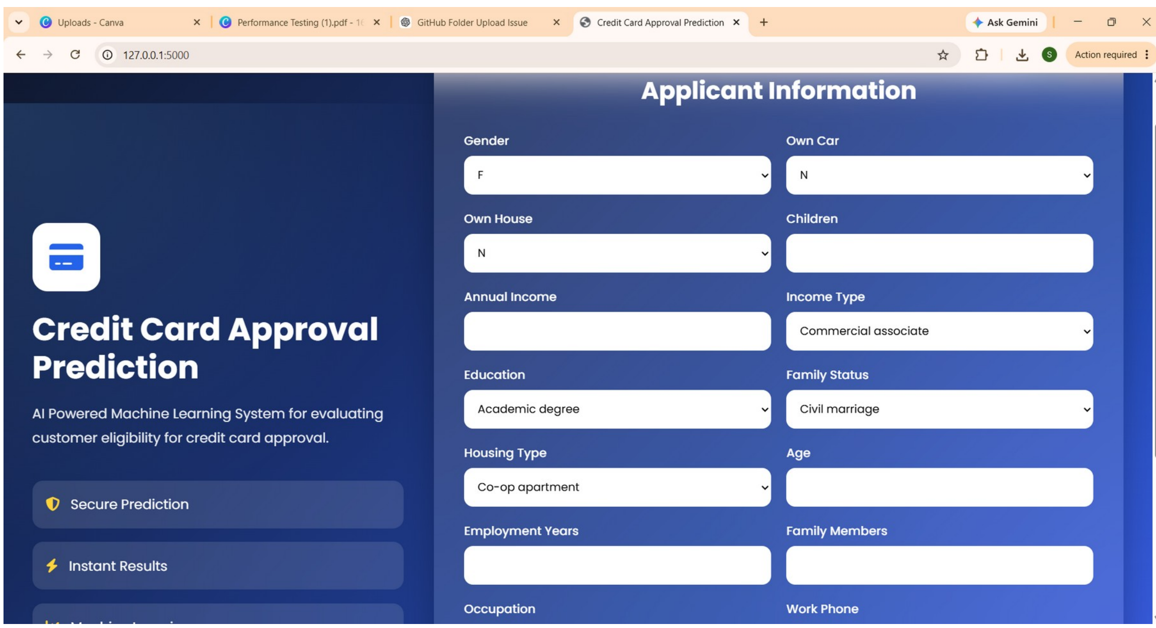

Figure 1: Applicant Information page with dynamically populated dropdowns

Predict Approval Button

After filling in all applicant fields, the user submits the form via the Predict Approval button, which sends the data to the /predict route.

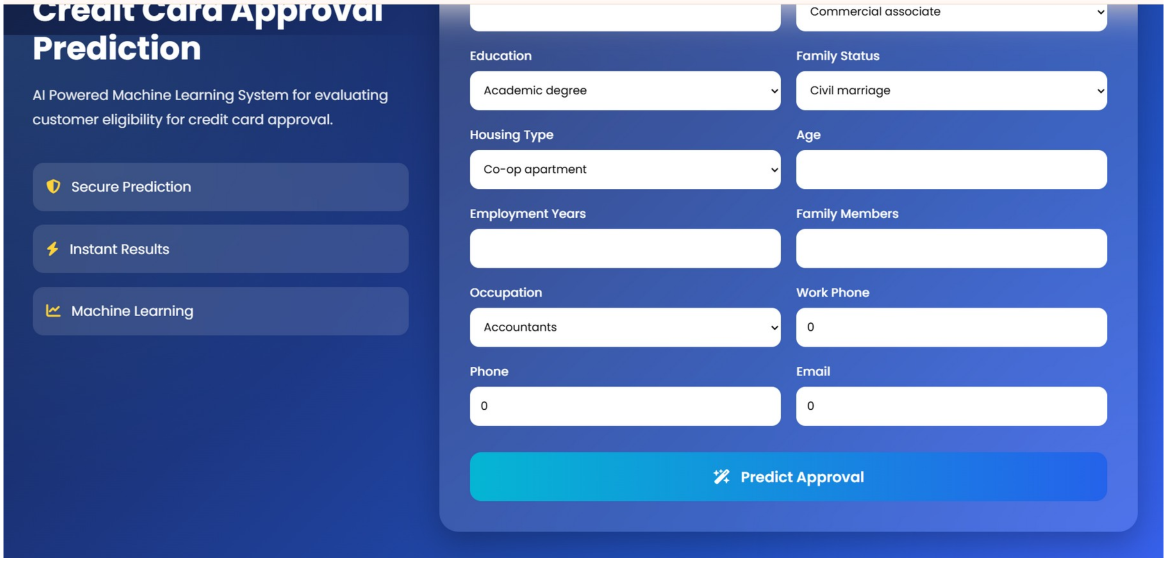

Figure 2: Completed form with the Predict Approval action

Approved Result

When the model predicts eligibility, the result page shows a green success state with a congratulatory message confirming the applicant meets the approval criteria.

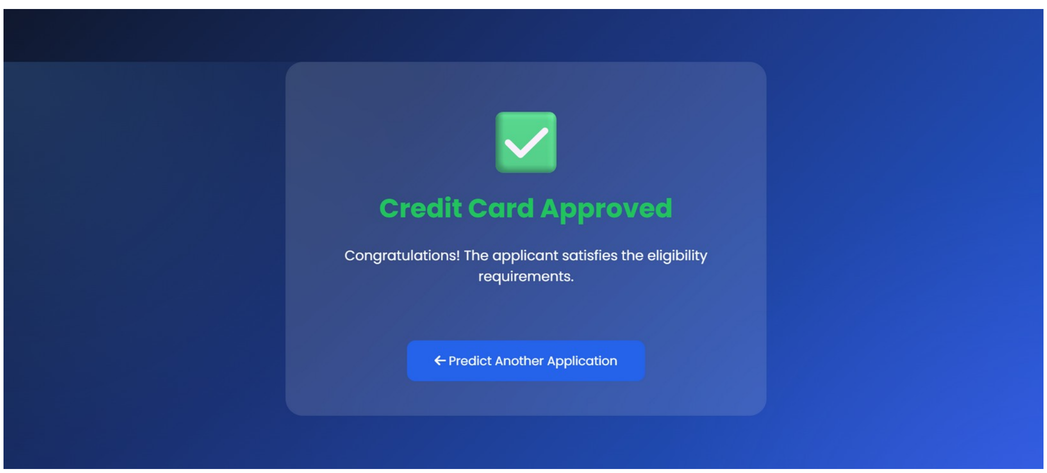

Figure 3: Approved prediction outcome

Rejected Result

When the model predicts a decline, the result page shows a red state with a clear rejection message and an option to try another application.

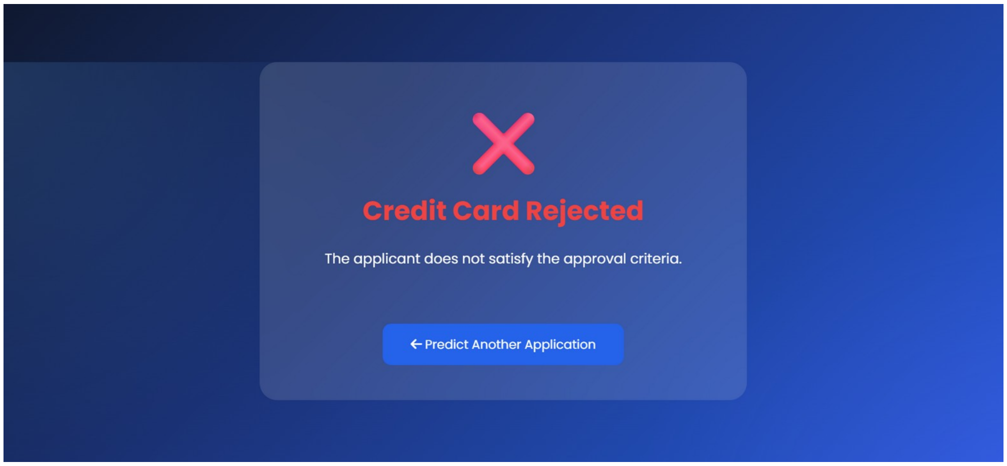

Figure 4: Rejected prediction outcome

Future Scope

- Set up CI/CD for automated deployments on Render.
- Improve prediction accuracy using deep learning models.
- Integrate real-time banking databases.
- Add secure user authentication.
- Develop a mobile-friendly interface.
- Enable online credit card application processing.
- Implement explainable AI for prediction transparency.

Team Member Details

S.No	Name	Role
1	Kodirekka Sangeeth Babu	Machine Learning & Flask Development
2	Aragonda Indra	Frontend Development & Testing

Conclusion

The Credit Card Approval Prediction System successfully demonstrates the application of machine learning in automating credit card approval decisions. By performing data preprocessing, feature engineering, model training, and evaluation, the system identifies the most suitable classification model for prediction. The trained model is integrated into a Flask web application, enabling users to obtain real-time approval or rejection results through an intuitive interface. This project reduces manual effort, improves consistency, and showcases the practical use of machine learning in banking and financial services.